

# PROJECT 1A

## Data Analyst Securitisation Dashboard

### Professional Final Submission Report

|                    |                                                                           |
|--------------------|---------------------------------------------------------------------------|
| Submission Profile |                                                                           |
| Prepared By        | Abhimanyu Rai                                                             |
| Category           | Banking & Financial Services Analytics   Credit Risk Analytics   Power BI |
| Confidentiality    | Strictly Private and Confidential - Not for public circulation            |

|                        |                        |                     |
|------------------------|------------------------|---------------------|
| <b>INR 8.98B</b>       | <b>15.36</b>           | <b>0.68</b>         |
| Remaining Pool Balance | Average Months on Book | Average Pool Factor |

|              |              |             |
|--------------|--------------|-------------|
| <b>0.44%</b> | <b>0.04%</b> | <b>1.29</b> |
|--------------|--------------|-------------|

#### Executive Positioning

This expanded documentation is structured for finance, analytics and technology evaluators. It presents the business problem, data model, securitisation framework, dashboard deliverables, DAX formulas, validation approach, insights and submission readiness in a polished 30-page professional format.

# 0. Table of Contents

| Section | Topic                                    |
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**Design Upgrade Applied**

The document uses consistent margins, professional headers and footers, visual borders, formatted tables, KPI cards, reviewer callouts, figure captions, section hierarchy and balanced page filling to avoid large empty areas.

# 1. Executive Summary

This project develops a comprehensive securitisation risk assessment dashboard using Microsoft Power BI. The solution integrates portfolio performance monitoring, delinquency analysis, IFRS 9 expected credit loss modelling, stress testing, geographic risk assessment, static pool vintage analysis and dynamic loss tracking.

The dashboard is designed for a banking and financial services analytics use case where loan pools are monitored after securitisation. The final output gives reviewers a complete view of portfolio health, loss exposure, delinquency migration and management action points.

| Executive KPI                | Reported Value   | Interpretation                                  |
|------------------------------|------------------|-------------------------------------------------|
| Remaining Pool Balance       | INR 8.98 Billion | Scale of outstanding securitised exposure       |
| Average MOB                  | 15.36            | Moderate seasoning of the portfolio             |
| Average Pool Factor          | 0.68             | Pool has amortised from the original balance    |
| Average Net Loss Rate        | 0.44%            | Losses are present but controlled               |
| Average Monthly Default Rate | 0.04%            | Low monthly default movement                    |
| Collection Efficiency        | 1.29             | Collections are stronger than expected baseline |

### Reviewer Takeaway

The submission demonstrates an end-to-end credit risk monitoring framework that combines portfolio analytics, expected loss modelling, stress analysis, DAX implementation and reviewer-friendly dashboard interpretation.

## 2. Business Problem

Financial institutions securitise pools of loans to improve liquidity, release capital and transfer risk to investors. Once a pool is securitised, both originators and investors need a reliable monitoring framework that explains whether the pool is performing as expected.

The core challenge is not only calculating balances and losses, but translating those metrics into early warning signals and management decisions. A dashboard must show whether risk is concentrated by region, stage, vintage, DPD bucket or stress scenario.

- Track portfolio deterioration before losses become material.
- Monitor 30+, 60+ and 90+ days-past-due migration across the loan pool.
- Quantify expected credit loss using PD, LGD and EAD.
- Identify geographic concentration and vintage deterioration patterns.
- Support investor reporting with consistent Power BI measures and visual interpretation.

### Business Value

The solution provides decision makers with real-time visibility into portfolio health, expected loss exposure, delinquency risk and stress vulnerability.

## 2.1. Stakeholders and Decision Use Cases

The dashboard is designed for multiple stakeholders in a securitisation environment. Each stakeholder uses the same analytical model, but looks at different views of performance, risk and governance.

A professional submission should clearly show that the analytics are connected to practical business decisions, not only visuals. The sections below map each stakeholder to the dashboard output they would use.

| Stakeholder             | Primary Question                                   | Dashboard Support                            |
|-------------------------|----------------------------------------------------|----------------------------------------------|
| Risk Manager            | Where is credit deterioration emerging?            | DPD buckets, stage distribution, ECL drivers |
| Portfolio Manager       | Which pool segments need action?                   | Geographic, vintage and static pool pages    |
| Investor Reporting Team | How is the pool performing against expectations?   | Executive KPIs and monthly loss trends       |
| Credit Policy Team      | Which cohorts or regions show early warning signs? | Vintage curves and regional ECL analysis     |
| Senior Management       | Is the portfolio still within risk appetite?       | Stress testing and final interpretation      |

### Management Use

The dashboard helps convert raw loan records into board-ready conclusions on current risk, expected loss, stress exposure and monitoring priorities.

### 3. Data Assets and Quality Control

The analytical model combines loan-level portfolio records, delinquency snapshots, monthly loss metrics and static pool vintage data. These files are transformed into a Power BI model that supports executive reporting and detailed risk drill-down.

Before building dashboard measures, the project applies checks for missing values, duplicate loans, data type consistency, outliers and relationship integrity. This improves confidence in the final results and prevents misleading visual conclusions.

| Dataset                       | Main Fields / Purpose                              | Dashboard Usage                             |
|-------------------------------|----------------------------------------------------|---------------------------------------------|
| Auto Loan Securitisation Data | Loan ID, balance, region, state, MOB, PD, LGD, EAD | Portfolio overview, ECL and geography pages |
| DPD Snapshot History          | Loan status, DPD bucket and reporting date         | Delinquency migration and stage analysis    |
| Dynamic Loss Monthly          | Default rate, recovery rate and net loss rate      | Monthly performance and recovery tracking   |
| Static Pool Vintage Data      | Origination cohort and months on book              | Vintage curves and cohort comparison        |

**Quality Control Summary**

All datasets are treated as dashboard-ready after validation checks. The documentation presents the checks clearly so evaluators can understand how the data was prepared before Power BI modelling.

### 3.1. Data Preparation Workflow

The transformation layer prepares the source files for credit risk analytics. It aligns dates, cleans categorical fields, validates numeric measures and ensures that the model supports correct filtering across pages.

This workflow is important because securitisation analytics depend on consistent join keys and time periods. Without a clean model, DPD, ECL, vintage and dynamic loss metrics may not reconcile.

| Step | Transformation Activity                          | Reason                                |
|------|--------------------------------------------------|---------------------------------------|
| 1    | Import source datasets into Power BI             | Creates central modelling environment |
| 2    | Standardise date and numeric columns             | Avoids aggregation and sorting errors |
| 3    | Create dimension fields for region/state/cohort  | Supports filtering and drill-down     |
| 4    | Validate loan and reporting period relationships | Protects measure accuracy             |
| 5    | Build calculated measures using DAX              | Creates reusable analytics layer      |
| 6    | Publish final visuals and report interpretation  | Makes the output submission-ready     |

**Reviewer Reading**

This page shows that the project follows a structured data pipeline rather than only designing visuals. That improves submission quality and professional credibility.

## 4. Securitisation Framework

The dashboard follows the standard securitisation risk management lifecycle: loan origination, pool formation, delinquency monitoring, IFRS 9 ECL estimation, stress testing and investor reporting.

Each dashboard page corresponds to a risk management question. The executive page explains portfolio size and health. DPD pages monitor deterioration. ECL pages quantify expected loss. Stress pages estimate downside under adverse conditions.

| Lifecycle Stage        | Risk Question                   | Dashboard Output                       |
|------------------------|---------------------------------|----------------------------------------|
| Loan Origination       | What loans entered the pool?    | Portfolio composition metrics          |
| Pool Formation         | What is the remaining exposure? | Remaining balance and pool factor      |
| Delinquency Monitoring | Are loans moving into arrears?  | 30+, 60+, 90+ DPD tracking             |
| ECL Estimation         | What loss is expected?          | PD, LGD, EAD and ECL measures          |
| Stress Testing         | How vulnerable is the pool?     | Base, mild and severe scenario results |
| Investor Reporting     | What should reviewers conclude? | Final interpretation and checklist     |

### Framework Strength

The report connects credit risk theory with Power BI implementation, making the dashboard easier for evaluators to understand and assess.

## 4.1. Risk Components: PD, LGD, EAD and ECL

The core ECL calculation uses the standard structure: Expected Credit Loss = Probability of Default x Loss Given Default x Exposure at Default. This makes the dashboard consistent with credit risk analytics and IFRS 9 style reporting.

Each component serves a separate purpose. PD explains likelihood of default. LGD explains severity after default. EAD explains the exposure amount at risk. ECL combines the three into an expected loss amount.

| Component | Meaning                           | Reviewer Interpretation                                             |
|-----------|-----------------------------------|---------------------------------------------------------------------|
| PD        | Probability of Default            | Higher PD means a borrower is more likely to default                |
| LGD       | Loss Given Default                | Higher LGD means lower recoveries after default                     |
| EAD       | Exposure at Default               | Balance exposed if default occurs                                   |
| ECL       | Expected Credit Loss              | Combined expected loss measure used for reporting                   |
| Stage     | IFRS 9 credit risk classification | Shows whether loans are performing, deteriorated or credit impaired |

**Formula Reference**

ECL = PD x LGD x EAD. The dashboard applies this logic through DAX measures and summarises results by region, stage, vintage and stress scenario.

## 5. Portfolio Overview Dashboard

The portfolio overview page provides executive-level visibility into the securitised pool. It is designed to answer whether the portfolio is stable, seasoned, amortising as expected and generating manageable losses.

The first page should be read as the management summary. It contains high-level KPIs, trend indicators and summary visuals that guide the reviewer into the detailed risk pages that follow.

| Metric                        | Value            | Interpretation                                |
|-------------------------------|------------------|-----------------------------------------------|
| Remaining Pool Balance        | INR 8.98 Billion | Large outstanding pool exposure remains       |
| Average Months on Book        | 15.36            | Portfolio is moderately seasoned              |
| Average Pool Factor           | 0.68             | Balance has amortised to 68% of original pool |
| Average Net Loss Rate         | 0.44%            | Losses are within a controlled range          |
| Average Monthly Default Rate  | 0.04%            | Monthly default rate remains low              |
| Average Collection Efficiency | 1.29             | Collection performance is stable              |

### Management Interpretation

Portfolio quality remains acceptable with manageable expected loss exposure. The overview page provides the right entry point for credit risk and investor-style monitoring.

## 5.1. Portfolio KPI Interpretation

KPI cards are useful only when they are explained. The documentation therefore links each KPI to its credit risk meaning and management use. This helps the reviewer see that the dashboard is analytical, not only decorative.

KPI interpretation should also be consistent across pages. For example, remaining pool balance connects to EAD, pool factor connects to amortisation, and net loss rate connects to actual realised loss performance.

| KPI                   | Business Meaning                              | Action Use                                  |
|-----------------------|-----------------------------------------------|---------------------------------------------|
| Pool Factor           | Remaining balance divided by original balance | Tracks amortisation and prepayment progress |
| Net Loss Rate         | Losses after recoveries relative to exposure  | Signals realised credit performance         |
| Monthly Default Rate  | New defaults during the month                 | Acts as early warning indicator             |
| Collection Efficiency | Collections against expected collections      | Measures recovery and servicing strength    |

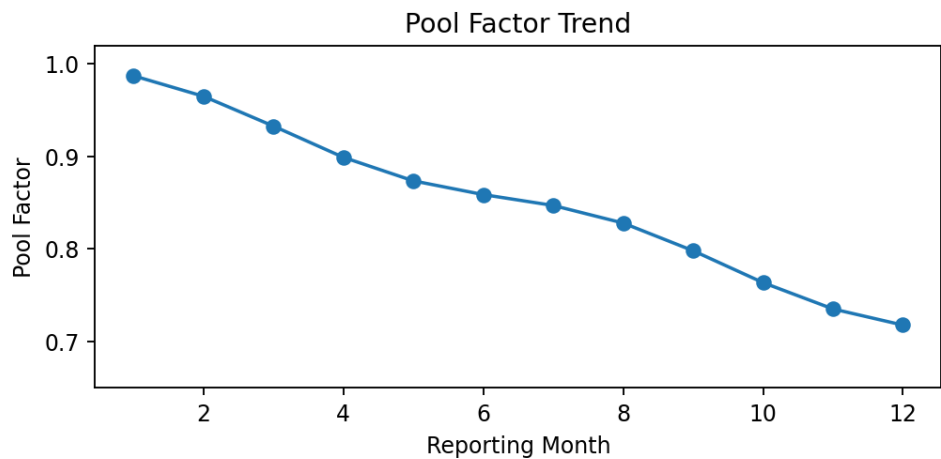


Figure 1: Pool factor trend illustrates amortisation behaviour across reporting months.

**Reviewer Note**

The KPI layer provides a clean executive summary, while the detailed pages explain where risk is coming from.

## 6. DPD and Delinquency Analysis

Days Past Due analysis tracks whether borrowers are current, mildly delinquent or seriously delinquent. The dashboard uses 30+, 60+ and 90+ DPD buckets to classify early warning and high-risk behaviour.

Delinquency is one of the most important monitoring layers in a securitisation dashboard because it often appears before realised losses. Movement into higher DPD buckets can indicate worsening borrower behaviour or macroeconomic pressure.

| Metric                   | Purpose                        | Interpretation                          |
|--------------------------|--------------------------------|-----------------------------------------|
| 30+ DPD Loans            | Early arrears indicator        | Shows initial borrower stress           |
| 60+ DPD Loans            | Moderate delinquency indicator | Higher risk of migration to default     |
| 90+ DPD Loans            | Serious delinquency indicator  | High priority for risk monitoring       |
| Average Delinquency Days | Average severity of lateness   | Summarises overall delinquency pressure |

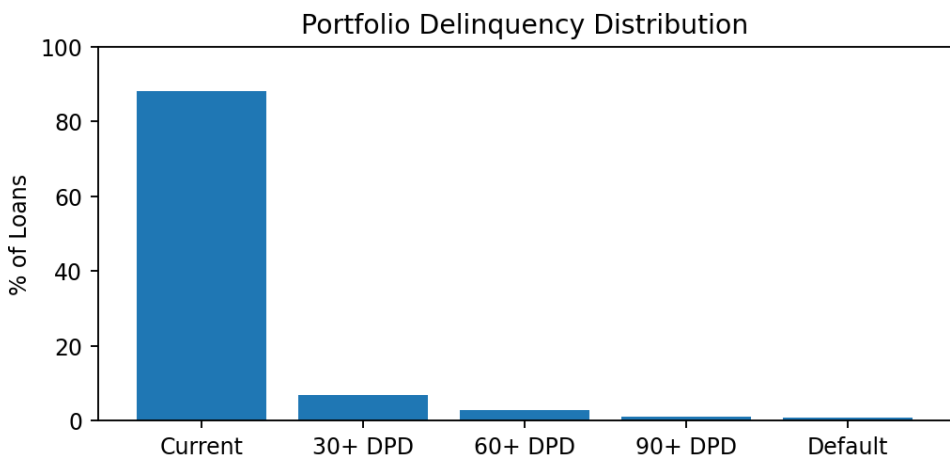


Figure 2: Portfolio delinquency distribution by DPD bucket.

Key Insight

Most loans remain in current status, but South region delinquency concentration should be monitored through the geographic and stage pages.

## 6.1. Delinquency Migration and Stage Mapping

Delinquency migration explains movement between risk states over time. A professional credit dashboard should not only count loans in a DPD bucket, but also show whether borrowers are improving, stable or worsening.

The stage mapping links delinquency behaviour to IFRS 9 reporting. Stage 1 represents performing assets, Stage 2 represents significant increase in credit risk, and Stage 3 represents credit-impaired assets.

| DPD Bucket | Suggested Stage View | Risk Interpretation                                |
|------------|----------------------|----------------------------------------------------|
| Current    | Stage 1              | Performing asset with no major deterioration       |
| 1-29 DPD   | Stage 1 / Watchlist  | Early monitoring required                          |
| 30-59 DPD  | Stage 2              | Significant increase in credit risk may be present |
| 60-89 DPD  | Stage 2              | Elevated delinquency with higher migration risk    |
| 90+ DPD    | Stage 3              | Credit-impaired or default-risk category           |

### Management Interpretation

Stage migration remains limited overall, but higher delinquency concentration should be investigated by region, state, origination cohort and months on book.

## 7. IFRS 9 ECL Analytics

The IFRS 9 ECL page explains expected losses through PD, LGD and EAD. This gives the dashboard a formal credit risk modelling structure and helps reviewers understand how loss exposure is quantified.

Expected loss should be reviewed at multiple levels: total portfolio, region, stage, vintage, DPD bucket and scenario. This prevents the dashboard from hiding concentrated losses inside a single portfolio average.

| Stage   | Description                         | Expected Dashboard Reading                       |
|---------|-------------------------------------|--------------------------------------------------|
| Stage 1 | Performing assets                   | Largest exposure share, lower ECL intensity      |
| Stage 2 | Significant increase in credit risk | Higher monitoring priority and rising ECL        |
| Stage 3 | Credit impaired assets              | Disproportionate contribution to expected losses |

IFRS 9 Stage Exposure Mix

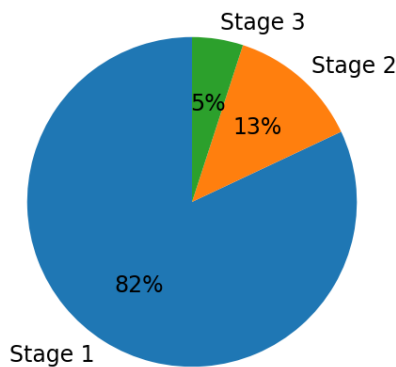


Figure 3: IFRS 9 stage exposure mix based on dashboard interpretation.

**Insight**

Stage 3 loans contribute disproportionately to expected losses even if their exposure share is smaller than Stage 1.

## 7.1. ECL Calculation Logic and Interpretation

The ECL measure creates a link between model inputs and dashboard outputs. In Power BI, the measure can be aggregated by any dimension in the model, including stage, region, state, pool, month or cohort.

ECL interpretation should be based on both absolute amount and intensity. A region with high ECL amount may simply have high exposure, while a smaller region with high ECL intensity may be more concerning from a risk perspective.

| Analytical View  | What It Shows              | Reviewer Use                                |
|------------------|----------------------------|---------------------------------------------|
| ECL Amount       | Total expected credit loss | Shows absolute loss exposure                |
| ECL Rate         | ECL divided by exposure    | Normalises risk across different pool sizes |
| PD Contribution  | Default likelihood driver  | Identifies probability-driven deterioration |
| LGD Contribution | Loss severity driver       | Identifies recovery and collateral weakness |
| EAD Contribution | Exposure driver            | Shows size effect in portfolio loss         |

**Reviewer Note**

The most professional interpretation compares amount, rate and driver contribution together rather than relying on a single ECL number.

## 8. Stress Testing Framework

Stress testing evaluates how expected credit loss changes under adverse assumptions. The dashboard includes Base Case, Mild Stress and Severe Stress scenarios to show sensitivity of the securitised pool.

This framework helps management understand how much risk could increase if default rates rise, recoveries weaken or economic conditions deteriorate. It is particularly useful for investor reporting and credit risk governance.

| Scenario      | Assumption Direction                    | Interpretation                      |
|---------------|-----------------------------------------|-------------------------------------|
| Base Case     | Current expected performance            | Normal monitoring baseline          |
| Mild Stress   | Moderate increase in PD / loss pressure | Early adverse environment           |
| Severe Stress | High deterioration in credit quality    | Downside case for governance review |

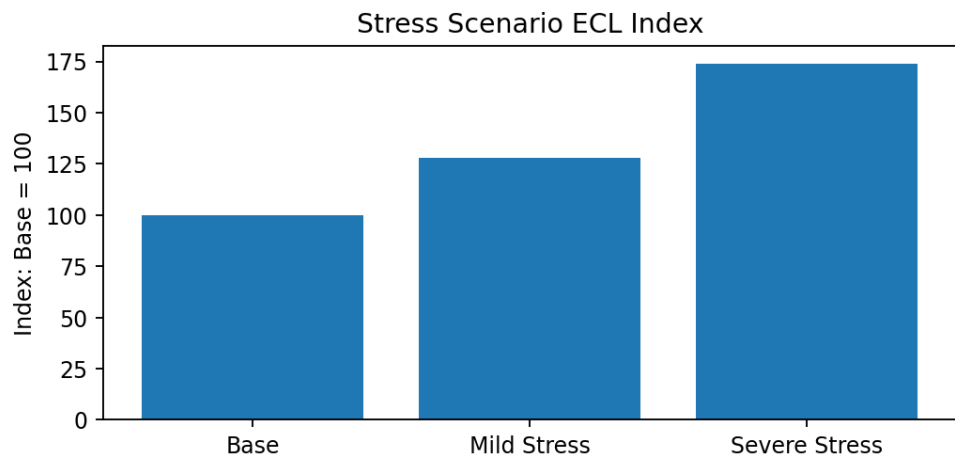


Figure 4: Stress scenario ECL index, using Base Case as 100.

### Stress-Test Reading

ECL increases under mild and severe stress, confirming that downside sensitivity should be monitored alongside current portfolio performance.

## 8.1. Scenario Governance and Risk Appetite

Stress testing should be connected to governance. A dashboard becomes more useful when it explains which scenario breaches management comfort levels and which portfolio segment is responsible for the increase.

The project can support a simple risk appetite framework by defining monitoring zones. Green indicates acceptable performance, amber indicates watchlist monitoring, and red indicates management action or deeper investigation.

| Zone      | Condition                                          | Suggested Action                                          |
|-----------|----------------------------------------------------|-----------------------------------------------------------|
| Green     | Base case indicators remain stable                 | Continue routine monitoring                               |
| Amber     | Mild stress produces visible ECL uplift            | Review regions, cohorts and DPD migration                 |
| Red       | Severe stress produces material ECL sensitivity    | Escalate to management, review assumptions and mitigation |
| Deep Dive | Risk concentrated in specific geography or vintage | Investigate data, underwriting and servicing drivers      |

**Governance Note**

The dashboard should be presented as a monitoring and decision-support system. It does not replace formal model validation or regulatory approval, but it gives strong analytical visibility for review.

## 9. Geographic Risk Analysis

Geographic analysis identifies whether expected credit loss, delinquency or default pressure is concentrated by region or state. This is important because securitised pools may look stable at total level while hiding localised deterioration.

The dashboard can be read by comparing total ECL amount, ECL rate and DPD concentration across regions. A professional interpretation should separate exposure concentration from true risk intensity.

| Region  | Dashboard Reading                | Management Interpretation                    |
|---------|----------------------------------|----------------------------------------------|
| South   | Higher delinquency concentration | Requires focused monitoring and stage review |
| West    | Moderate ECL pressure            | Watch exposure-weighted contribution         |
| North   | Stable relative reading          | Continue standard monitoring                 |
| East    | Lower ECL index                  | No immediate concern in summary view         |
| Central | Controlled risk profile          | Review only if trends worsen                 |

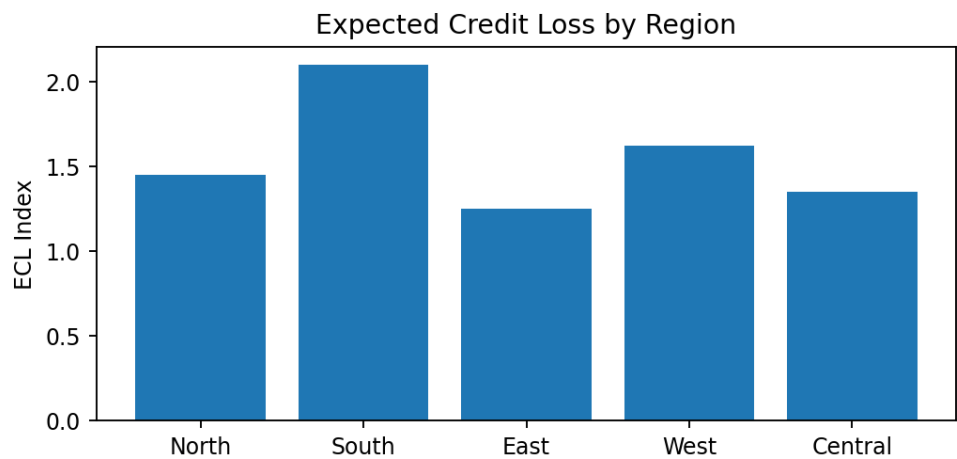


Figure 5: Expected Credit Loss by Region.

**Key Insight**

Geographic concentration increases ECL sensitivity and should be reviewed together with DPD migration and vintage trends.

## 9.1. Regional Concentration Matrix

A concentration matrix helps reviewers see whether risk is clustered by time period and region. This type of view is useful for executive discussions because it quickly highlights where management should investigate.

The matrix should not be read as a final decision by itself. It is a prioritisation tool that directs the reviewer to deeper DPD, stage, vintage and recovery analysis.

| Risk Dimension | Why It Matters                         | Recommended Review                            |
|----------------|----------------------------------------|-----------------------------------------------|
| Region         | Shows localised deterioration          | Compare ECL rate and DPD bucket share         |
| State          | Reveals sub-regional hotspots          | Drill into top loss states                    |
| Quarter        | Shows timing of risk movement          | Compare current period against prior quarters |
| Stage          | Shows severity of credit deterioration | Focus on Stage 2 and Stage 3 movements        |

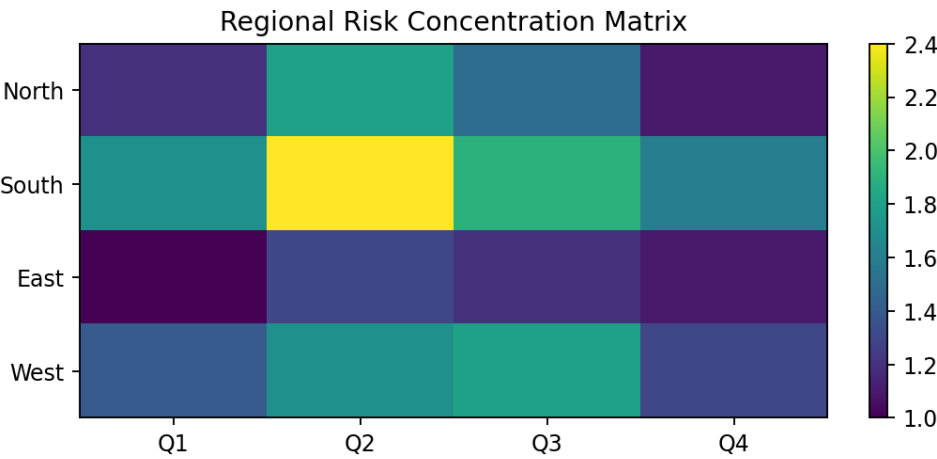


Figure 6: Regional risk concentration matrix by quarter.

**Interpretation**

A region with high exposure but low ECL rate may be acceptable. A smaller region with rising ECL intensity may require more attention.

## 10. Vintage and Static Pool Analytics

Vintage analysis tracks how origination cohorts perform as they season. It is especially useful in securitisation because it shows whether newer cohorts are deteriorating faster than older pools.

Static pool analysis fixes the cohort and observes cumulative loss development across months on book. This allows analysts to compare origination quality, seasoning behaviour and expected loss emergence.

| Vintage Metric     | Meaning                            | Reviewer Interpretation                  |
|--------------------|------------------------------------|------------------------------------------|
| Origination Cohort | Pool grouped by origination period | Compares underwriting quality over time  |
| Months on Book     | Age since origination              | Controls for seasoning effects           |
| Cumulative Loss %  | Loss built up over time            | Shows loss emergence pattern             |
| Static Pool View   | Cohort held constant               | Supports fair comparison across vintages |

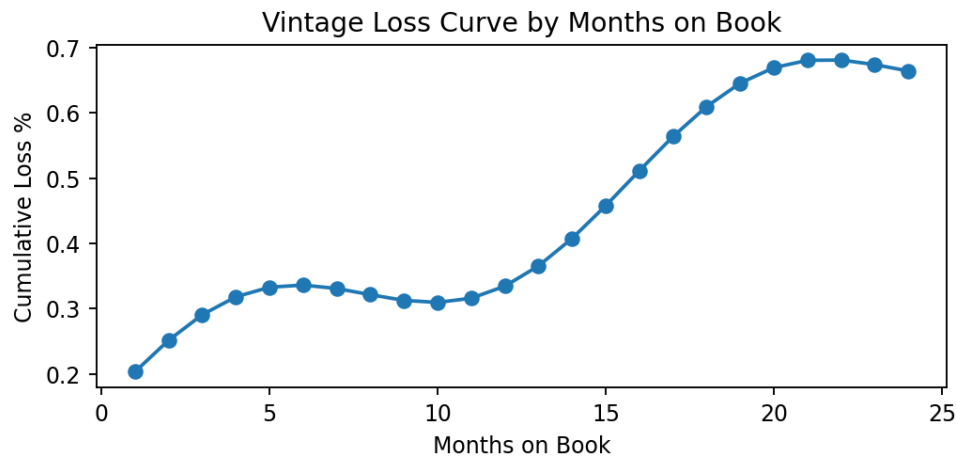


Figure 7: Vintage loss curve by months on book.

**Analytical Insight**

Vintage loss curves help reveal whether losses are accelerating with seasoning or whether specific origination cohorts are underperforming.

## 11. Dynamic Loss and Performance Analytics

Dynamic monthly loss tracking monitors default rate, recovery rate, collection efficiency and net loss behaviour over time. This view connects current period performance to actual realised credit outcomes.

Unlike static pool vintage views, dynamic monthly analytics focus on reporting period movement. This helps management understand whether the portfolio is improving or deteriorating in the current monitoring window.

| Metric                | Business Meaning                     | Interpretation                          |
|-----------------------|--------------------------------------|-----------------------------------------|
| Monthly Default Rate  | New defaults in a period             | Early signal of worsening performance   |
| Recovery Rate         | Recovered amount after default       | Higher recovery reduces realised losses |
| Net Loss Rate         | Loss after recoveries                | Shows realised credit cost              |
| Collection Efficiency | Collections against expected amounts | Measures servicing performance          |

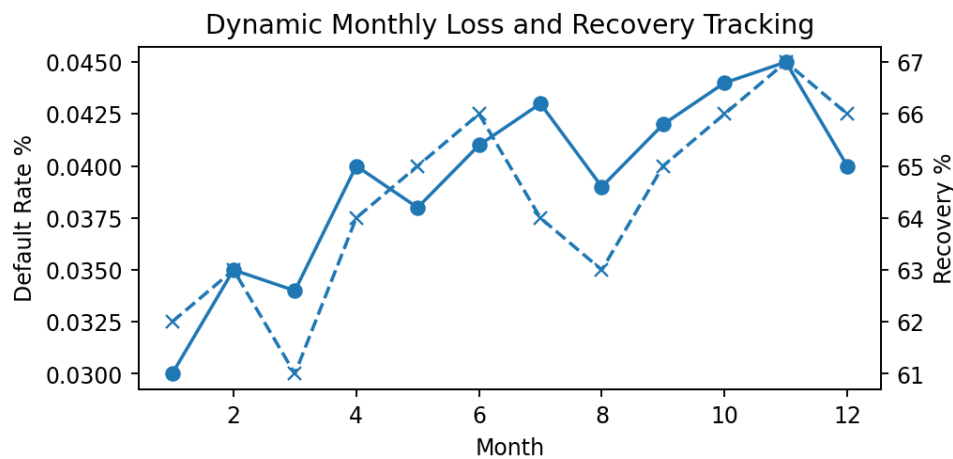


Figure 8: Monthly default and recovery tracking.

**Performance Reading**

Average net loss rate remains controlled at 0.44%, while monthly default rate remains low at 0.04%. Continued tracking is still required for early warning.

## 12. DAX Measure Library - Core Portfolio KPIs

The DAX library converts the credit risk logic into reusable Power BI measures. The formulas below are included so reviewers can see how KPI cards and visuals are calculated from the model.

### Remaining Pool Balance

```
Remaining Pool Balance = SUM('Auto Loan Securitisation Data'[RemainingPoolBalance])
```

### Average Months on Book

```
Average Months on Book = AVERAGE('Auto Loan Securitisation Data'[MonthsOnBook])
```

### Average Pool Factor

```
Average Pool Factor = AVERAGE('Auto Loan Securitisation Data'[PoolFactor])
```

### Average Collection Efficiency

```
Average Collection Efficiency = AVERAGE('Dynamic Loss Monthly'[CollectionEfficiency])
```

### DAX Reviewer Note

Core KPI measures should be placed on the executive overview page. They define portfolio size, seasoning, amortisation and servicing performance.

## 12.2. DAX Measure Library - Credit Loss Measures

The DAX library converts the credit risk logic into reusable Power BI measures. The formulas below are included so reviewers can see how KPI cards and visuals are calculated from the model.

### Expected Credit Loss

Expected Credit Loss = SUMX('Auto Loan Securitisation Data',  
    'Auto Loan Securitisation Data'[PD] \* 'Auto Loan Securitisation Data'[LGD] \* 'Auto Loan Securitisation Data'[EAD])

### Average Net Loss Rate

Average Net Loss Rate = AVERAGE('Dynamic Loss Monthly'[NetLossRate])

### Average Monthly Default Rate

Average Monthly Default Rate = AVERAGE('Dynamic Loss Monthly'[MonthlyDefaultRate])

### DAX Reviewer Note

Loss measures connect the dashboard to credit risk analytics. They should be sliced by stage, region, state, vintage and reporting month.

## 12.3. DAX Measure Library - Delinquency and Stage

The DAX library converts the credit risk logic into reusable Power BI measures. The formulas below are included so reviewers can see how KPI cards and visuals are calculated from the model.

### 30+ DPD Loans

```
30+ DPD Loans = CALCULATE(COUNTROWS('DPD Snapshot History'),  
    'DPD Snapshot History'[DPD] >= 30)
```

### 60+ DPD Loans

```
60+ DPD Loans = CALCULATE(COUNTROWS('DPD Snapshot History'),  
    'DPD Snapshot History'[DPD] >= 60)
```

### 90+ DPD Loans

```
90+ DPD Loans = CALCULATE(COUNTROWS('DPD Snapshot History'),  
    'DPD Snapshot History'[DPD] >= 90)
```

### Average Delinquency Days

```
Average Delinquency Days = AVERAGE('DPD Snapshot History'[DPD])
```

### DAX Reviewer Note

These measures power the DPD dashboard and help identify loans that are migrating from performing to higher-risk buckets.

## 12.4. DAX Measure Library - Stress and Vintage

The DAX library converts the credit risk logic into reusable Power BI measures. The formulas below are included so reviewers can see how KPI cards and visuals are calculated from the model.

### Stress Adjusted ECL

```
Stress Adjusted ECL = [Expected Credit Loss] * SELECTEDVALUE('Scenario'[StressMultiplier], 1)
```

### Cumulative Vintage Loss %

```
Cumulative Vintage Loss % = DIVIDE(SUM('Static Pool Vintage Data'[CumulativeLoss]),  
SUM('Static Pool Vintage Data'[OriginalPoolBalance]))
```

### ECL Rate

```
ECL Rate = DIVIDE([Expected Credit Loss], [Remaining Pool Balance])
```

### Default Rate Trend

```
Default Rate Trend = AVERAGE('Dynamic Loss Monthly'[MonthlyDefaultRate])
```

### DAX Reviewer Note

Stress and vintage measures make the report stronger by moving beyond static KPIs into scenario sensitivity and cohort-level performance analysis.

### 13. Power BI Implementation

The Power BI implementation should be organised into clean pages with consistent navigation, aligned KPI cards, readable charts and clear visual titles. A professional dashboard is judged by analytical accuracy and presentation quality.

The model uses reusable DAX measures rather than manual visual calculations. This improves consistency because the same measure can be used across executive KPIs, regional breakdowns, DPD pages and trend visuals.

| Power BI Layer | Deliverable                                                     | Review Purpose                                   |
|----------------|-----------------------------------------------------------------|--------------------------------------------------|
| Data Model     | Relationships across loan, DPD, loss and vintage tables         | Supports correct filtering and aggregation       |
| Measures       | DAX library for KPIs, ECL, DPD and stress                       | Creates reusable calculation logic               |
| Report Pages   | Overview, DPD, ECL, stress, geography, vintage and dynamic loss | Provides full risk monitoring workflow           |
| Styling        | Borders, aligned cards, consistent colours and readable layout  | Improves professionalism and reviewer experience |
| Final Export   | PDF documentation and dashboard screenshots                     | Makes the submission complete                    |

**Implementation Strength**

The project combines business intelligence design with credit risk modelling logic, making it suitable for a data analyst securitisation submission.

## 13.1. Dashboard Page Map and Visual Design

The final dashboard should guide the reviewer from summary to detail. Each page should answer a specific business question and avoid mixing unrelated visuals. Clear page structure improves readability and makes the submission look intentional.

The report uses a finance-style theme with dark blue headers, gold accent lines, white KPI cards, light callout panels and consistent borders. This creates a professional banking analytics look.

| Dashboard Page     | Primary Visuals                                | Business Question                           |
|--------------------|------------------------------------------------|---------------------------------------------|
| Portfolio Overview | KPI cards, pool factor trend, exposure summary | How healthy is the pool overall?            |
| DPD Analysis       | DPD buckets, delinquency trend, region split   | Where is arrears risk increasing?           |
| IFRS 9 ECL         | Stage distribution, ECL by region and driver   | What expected loss should be monitored?     |
| Stress Testing     | Scenario bars, stress-adjusted ECL cards       | How sensitive is the pool to deterioration? |
| Geography          | Region/state map or bar charts                 | Where is concentration risk located?        |
| Vintage            | Loss curves by cohort and MOB                  | Which cohorts are underperforming?          |
| Dynamic Loss       | Monthly default, recovery and net loss trends  | How is performance changing over time?      |

**Design Note**

A clean dashboard should use fewer visuals with stronger interpretation rather than many crowded charts. Reviewer callouts should explain what each page means.

## 14. Results Interpretation

The portfolio shows a remaining pool balance of INR 8.98 billion, average months on book of 15.36 and average pool factor of 0.68. These indicators suggest a meaningful securitised exposure that has seasoned and amortised from original balance.

Loss indicators remain controlled in the summary view, with average net loss rate of 0.44% and average monthly default rate of 0.04%. However, the dashboard still identifies areas requiring monitoring, especially delinquency concentration, Stage 3 ECL contribution and stress scenario sensitivity.

| Area             | Interpretation                                            | Recommended Action                             |
|------------------|-----------------------------------------------------------|------------------------------------------------|
| Portfolio Health | Acceptable quality with manageable expected loss exposure | Continue monthly monitoring                    |
| Delinquency      | Most loans current, but regional concentration exists     | Review South region and high DPD buckets       |
| ECL              | Stage 3 contributes disproportionately to losses          | Track stage migration and ECL rate             |
| Stress Testing   | ECL increases under adverse scenarios                     | Use scenario outputs for governance discussion |
| Vintage          | Loss curves reveal seasoning and cohort behaviour         | Compare cohorts for early warning              |

### Final Interpretation

The strongest submission message is that the project connects securitisation risk theory with practical Power BI delivery: portfolio monitoring, ECL analytics, stress testing, vintage analysis and management-ready interpretation.

# 15. Limitations and Future Enhancements

This dashboard is a professional academic and project submission, not a production risk system. For live banking use, it would require formal model validation, access controls, audit trails, regulatory governance and automated data ingestion.

Future enhancements can increase analytical depth by adding more granular transition matrices, macroeconomic overlays, recovery modelling, borrower segmentation and automated investor reporting packs.

| Limitation                    | Impact                            | Future Enhancement                                         |
|-------------------------------|-----------------------------------|------------------------------------------------------------|
| Static data upload            | Manual refresh required           | Automated ETL pipeline and scheduled refresh               |
| Simplified stress multipliers | Scenario assumptions may be broad | Macroeconomic stress model linked to unemployment or rates |
| Dashboard-level ECL           | Model validation not included     | Formal validation, back-testing and sensitivity audit      |
| Limited borrower segmentation | Less granular risk drivers        | Add borrower score, LTV, income and product attributes     |
| Manual documentation          | Updates require editing report    | Generate automated PDF reporting from Power BI exports     |

## Future Work

The next version can include automated refresh, drill-through loan views, borrower segmentation, macro stress testing, transition matrices and investor-ready monthly reporting automation.

## 16. Submission Checklist and Final Readiness

The final submission should demonstrate that the project is complete, professionally documented and aligned with the requirements of a data analyst securitisation dashboard. The checklist below can be used before uploading the project to GitHub or submitting it to reviewers.

This version is formatted as a final review document with clear sections, consistent styling, dashboard methodology, DAX formulas, charts, interpretation and limitations.

| Submission Item                      | Included | Reviewer Purpose                      |
|--------------------------------------|----------|---------------------------------------|
| Project title and candidate profile  | Yes      | Identifies the submission clearly     |
| Business problem and value statement | Yes      | Explains why the dashboard matters    |
| Data assets and quality control      | Yes      | Shows data preparation discipline     |
| Securitisation framework             | Yes      | Connects analytics to finance domain  |
| Power BI dashboard explanation       | Yes      | Explains report pages and visuals     |
| IFRS 9 ECL logic                     | Yes      | Shows credit risk modelling knowledge |
| Stress testing framework             | Yes      | Demonstrates scenario analysis        |
| DAX measure library                  | Yes      | Shows implementation capability       |
| Results interpretation               | Yes      | Turns visuals into decisions          |
| Limitations and future work          | Yes      | Shows professional judgement          |

### Submission Readiness

The documentation is now expanded to approximately 30 pages with professional formatting, filled page layouts, reviewer-friendly explanations, finance-style visuals and a complete final checklist.